GlucoTwin Bharat

A live, privacy-first digital twin for Type 2 diabetes in Indian patients. It predicts glucose spikes and lows 2 hours ahead, notices when the patient stops matching their record, and lets the doctor test a plan before prescribing it.

Team **Because I'm Happy Clap Along**

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The problem

India has over 100 million people with diabetes, but doctors see them every few months. Between visits they are blind to what actually happens.

Invisible spikes

High-GI Indian meals (rice, biryani, dosa) and late dinners after poor sleep drive repeated post-meal spikes no one sees in time.

Dangerous lows

Sulfonylureas plus fasting days (vrat) cause hypoglycaemia, especially in elderly patients.

Silent change

Infection, missed doses or stress shift a patient's physiology; the record and the doctor don't know until the next HbA1c.

Black-box tools don't help

Generic predictors trained on foreign data can't explain *why*, and can't answer "what if he ate millet instead?"

Privacy is a hard constraint

The DPDP Act makes cloud-heavy AI on identifiable patient data hard to deploy.

Our solution: a twin, not just a predictor

Two data streams fused into one patient-specific twin, read by a team of experts. No model trained from scratch.

Built from the record

HOMA-IR/HOMA-B, medicines and TCF7L2 genotype set the parameters of a published glucose–insulin physiology model. Works from day one.

Detects change

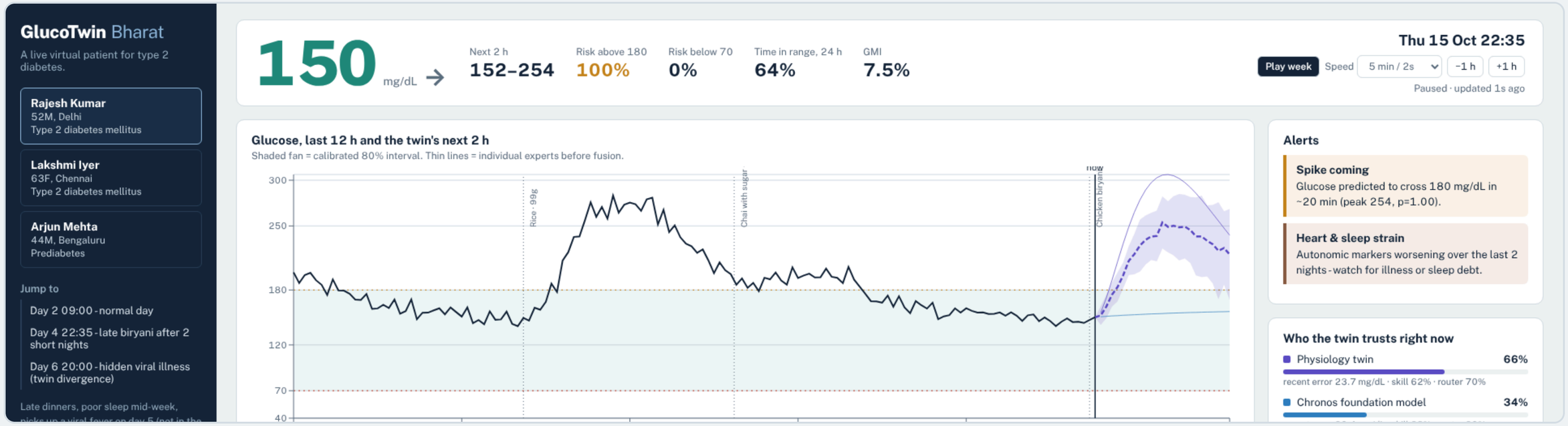
Every 2 h the twin re-fits itself to CGM. A sustained gap between patient and record raises a twin divergence alert.

Answers what-if

Replays the next 4 h under two plans (diet, walk, drug, sleep) so the doctor can test before prescribing.

Explains itself

An LLM router shows which expert it trusts and why; every alert carries a calibrated probability.



Architecture

Mixture of zero-shot experts → skill-weighted, LLM-routed fusion → calibrated alerts → doctor dashboard. Full diagram: GlucoTwin_Architecture.pdf

Inputs

- EHR: labs, diagnoses, meds, BP, genetics
- Wearables: CGM, HR, steps, sleep, raw PPG, meal log

Experts

- Physiology twin + data assimilation
- Amazon Chronos-Bolt (zero-shot)
- PaPaGei-S PPG model + HRV
- Framingham, IDRS, CGM metrics

Fusion & output

- $1/MAE^2$ skill weights on 8 h of hindcasts
- LLM router with sanity guard
- Split-conformal 80% band
- P(>180), P(<70) alerts

Edge & privacy

- All models on device
- Privacy gateway before any LLM
- Local LLM option: zero egress

Learning

- On-device online adapter (192 weights, RLS)
- Federated averaging + differential privacy

Stack

- Python, FastAPI, PyTorch
- React, Vite, Recharts
- DeepInfra Llama-3.1-8B / Ollama

Ask the twin: what if?



"Swap to millet roti and walk 20 minutes after." The twin rewinds to before the meal and simulates 4 hours both ways.

	Biryani	Millet + walk
Peak (mg/dL)	276	**191**
Minutes > 180	175	**95**
Glucose load	304	**124**

A forecasting model can't answer this: it has never seen this patient eat millet. A physiology simulator can.

Grounded chat and clinical brief



The assistant gets the live twin as context: record, glucose, forecast, alerts, sleep, meals and risk. What-if questions run the simulator first, then the LLM explains.

One-click brief

SUMMARY, ALERTS and three concrete, Indian-diet-aware ACTIONS, written only from the twin's own numbers.

Adverse events caught



Hidden illness

Rajesh's viral fever isn't in his record. The twin sees basal glucose ~40 mg/dL above his record and raises twin divergence.



Hypoglycaemia

Lakshmi on glimepiride, fasting day + temple walk: glucose 56, no recovery predicted; sulfonylurea flagged.



Cardiovascular strain

Arjun's HRV, computed from raw PPG, falls night after night while resting HR climbs → sustained autonomic strain flag.

Outcomes

Synthetic cohort of 23 patients, 3,481 two-hour forecasts, zero training. Synthetic data as required by the challenge sandbox rules.

0.95
AUROC for a spike >180 within 2 h of a logged meal (precision 0.98, recall 0.86)

35 min
Median warning before glucose crosses 180

0.97
AUROC for hypoglycaemia <70 within 2 h (recall 0.90)

28
mg/dL post-meal error vs 76 for a curve-only model

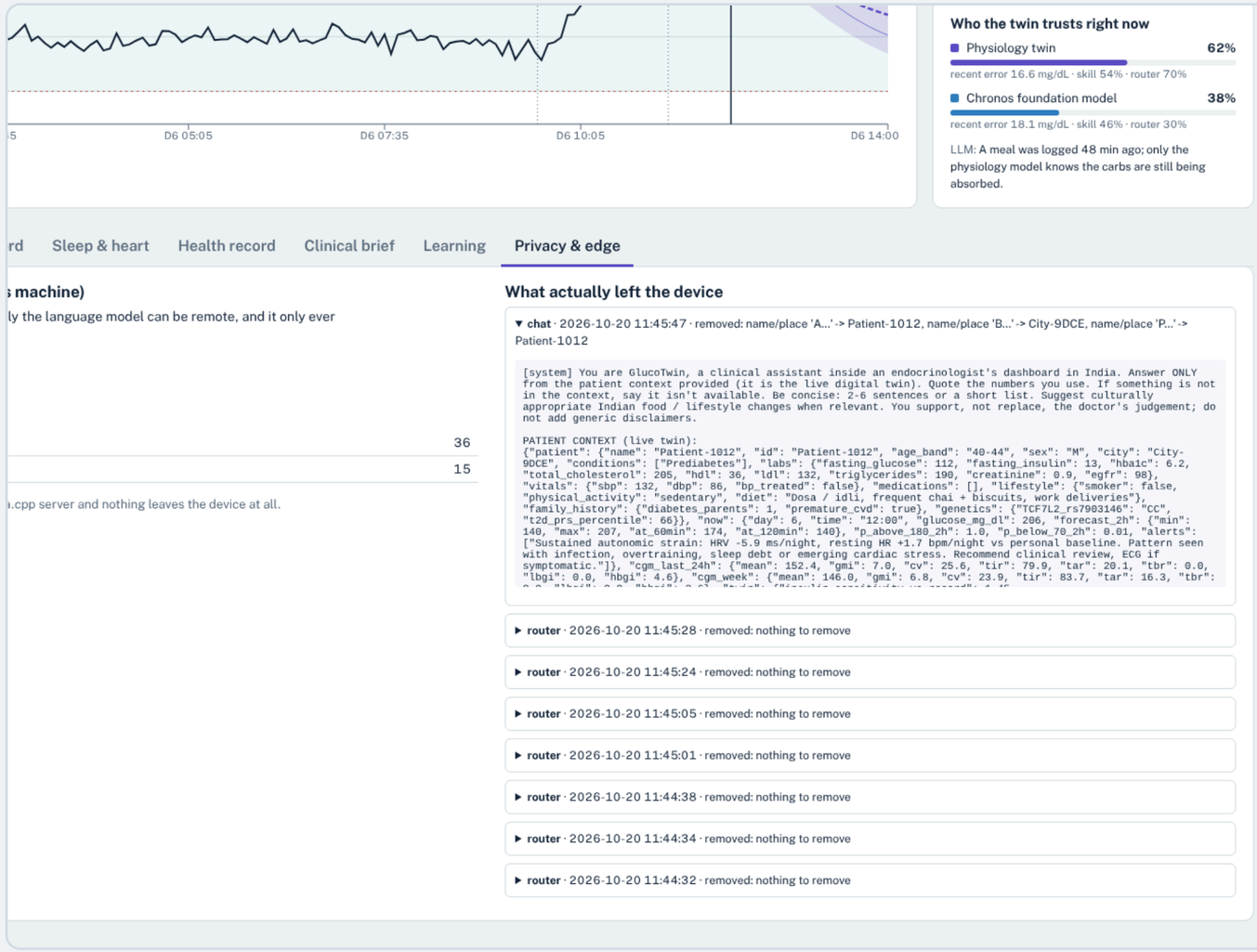
Forecast error (MAE, mg/dL)

Model	30 min	60 min	120 min
Persistence	18.4	31.0	51.3
Physiology twin	15.7	19.0	33.4
Fused twin	**11.6**	**17.6**	**33.6**

Learning & divergence

- Federated prior on unseen patients: MAE 16.6 → 15.8
- On-device adapter for Rajesh: recent error 27 → 20 mg/dL
- Hidden illness: basal shift −2.5 → +29.5 mg/dL after onset

Privacy and edge by design



Everything runs on the device

Physiology twin, Chronos, PaPaGei, adapter, alerts and simulator.

Privacy gateway

Names → pseudonyms; Aadhaar, ABHA, PAN, phone, email removed; age → band. Every outgoing payload is shown and logged.

Federated, not centralised

Only 192 clipped, noised weights leave a phone. Point the LLM at Ollama and nothing leaves at all.

Impact and roadmap

- Doctors see risk 2 hours ahead, with reasons
- Diet changes tested on the twin before the patient
- Illness and missed doses surfaced from glucose alone
- Runs on a laptop with open models; under a cent per session
- DPDP-compliant by design

- Validate on real CGM datasets (OhioT1DM, Shanghai T2DM)
- LoRA adapters on Chronos / PaPaGei with FedAvg
- Real-device federation (Flower) with formal DP accounting
- Patient-facing mobile app with nudges
- Synthea EHR + ABDM / ABHA integration

Demo video: youtu.be/j5wbkDxeSIs · Code: public GitHub repository (MIT License)